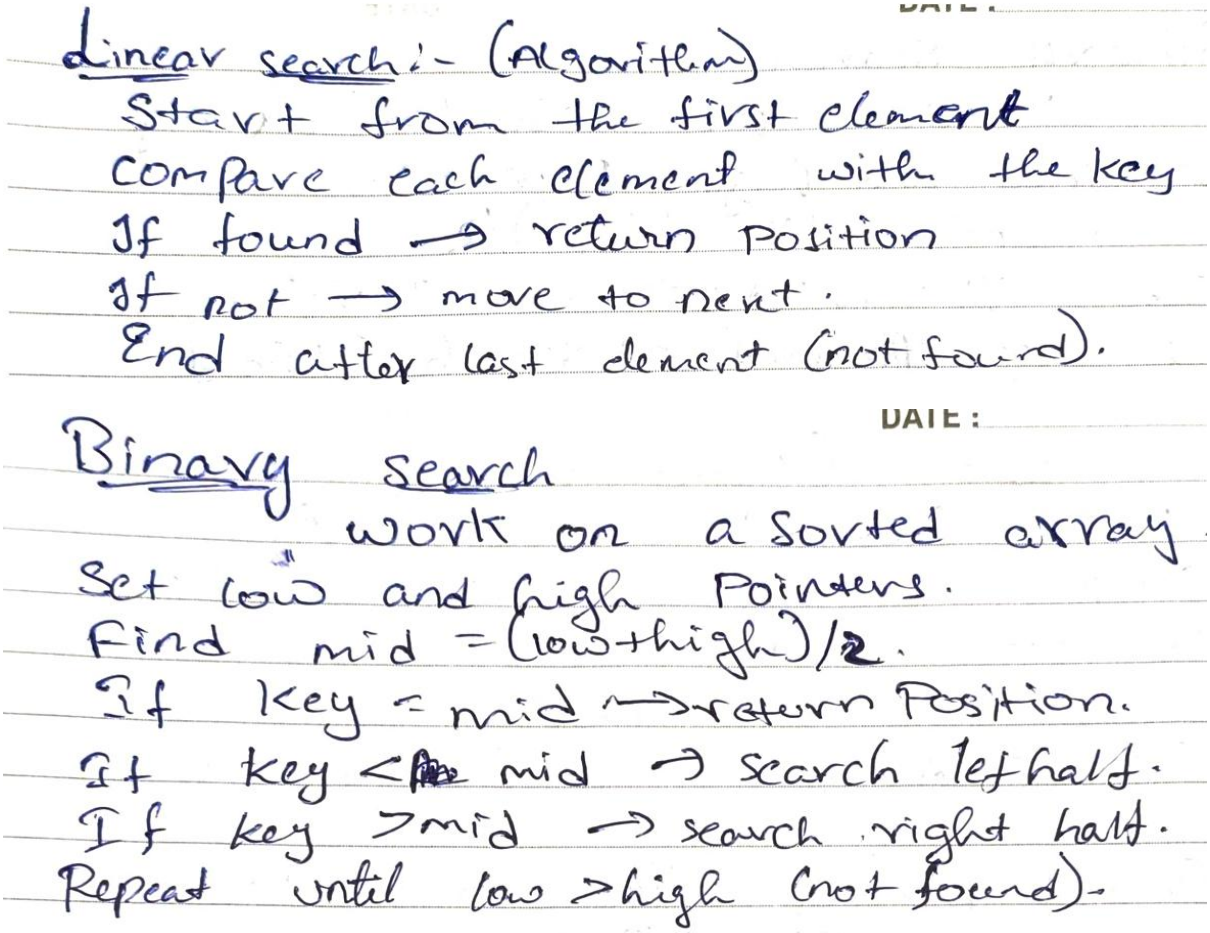
Practical 2

AIM: Implementation and Time analysis of linear and binary search algorithm.

1 ALGORITHM:

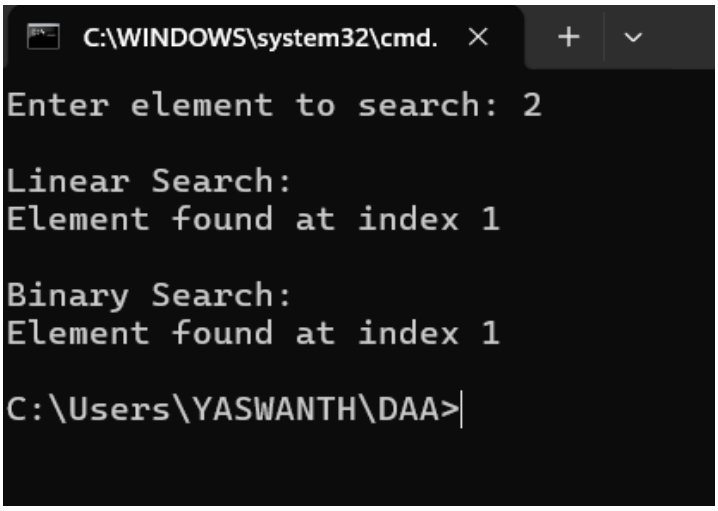


2 PROGRAM:

```
#include <iostream>
#include <vector>
using namespace std;

// Linear Search Function
int linearSearch(vector<int> arr, int key) {
    for (int i = 0; i < arr.size(); i++) {
        if (arr[i] == key)
            return i;
    }
    return -1;
}

// Binary Search Function
int binarySearch(vector<int> arr, int key) {
```

```
    int low = 0;
    int high = arr.size() - 1;
    while (low <= high) {
        int mid = low + (high - low) / 2;
        if (arr[mid] == key)
            return mid;
        else if (arr[mid] < key)
            low = mid + 1;
        else
            high = mid - 1;
    }
    return -1;
}

int main() {
    int n = 100;   // smaller size for easy testing
    vector<int> arr(n);
    // Creating a sorted array
    for (int i = 0; i < n; i++)
 arr[i] = i + 1;

    int key;
    cout << "Enter element to search: ";
    cin >> key;
    int index;
    // Linear Search
    index = linearSearch(arr, key);
    cout << "\nLinear Search:\n";
    if (index != -1)
        cout << "Element found at index " << index << endl;
    else
        cout << "Element not found\n";
    // Binary Search
    index = binarySearch(arr, key);
    cout << "\nBinary Search:\n";
    if (index != -1)
        cout << "Element found at index " << index << endl;
    else
        cout << "Element not found\n";
    return 0;
}
```

3 OUTPUT:



4 TIME COMPLEXITY:

Time complexity:- time to search in array size n.

$T(n) = T(n-1) + O(1)$.

Expand:-

$T(n) = T(n-1)+1 = T(n-2)+1+1 = \cdots T(1)+(n-1)$

Best case: $T(1) = O(1)$.

final:- $T(n) = O(n)$.

Best case: $O(1)$
Worst case: $O(n)$,
Average case: $O(n)$

Time complexity:-
Let $T(n)$ = time to search in array of size n.
for checking middle element content $O(1)$.
Search half the array $\rightarrow n/2$.

$T(n) = T(\frac{n}{2})+1 = T(\frac{n}{4})+1+1 = \cdots T(1)+\log n$.

Best case: $T(1) = O(1)$.

$T(n) = O(\log n)$.

Best Case = $O(1)$
Average case : $O(\log n)$
Worst case: $O(\log n)$.

5 Conclusion: LINEAR SEARCH HAS A TIME COMPLEXITY OF O(N), WHILE BINARY SEARCH RUNS IN O(LOG N) ON SORTED ARRAYS.